BMBA ORIGINAL SAVOLLARI

2025-yil 5-variant

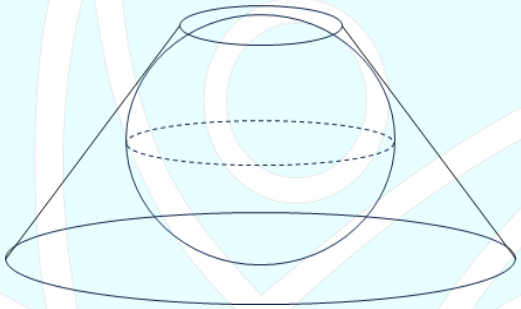
1. Agar $tga = 2$ bo'lsa, $\frac{3sina+2cosa}{2sina-3cosa}$ ni toping?

A) 1 B) 2 C) 5 D) 8

2. $F(x) = \log_7 x + ln^3 x$ funksiya berilgan. $F'(1)$ nuqtadagi qiymatini toping?

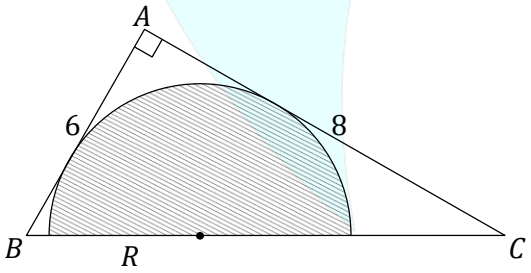
A) $\frac{1}{ln7}$ B) $\frac{1}{ln5}$ C) $\frac{3}{ln2}$ D) 1

3. Kesik konusga radiusi 3 cm ga teng bo'lgan shar ichki chizilgan. Kesik konusning kichik asosining radiusi yasovchisidan 4 marta kichik bo'lsa, kesik konus hajmini toping.



A) 64π B) $66\,\pi$ C) 78π D) 68π

4. Berilgan ma'lumotlardan foydalanib yarim doiraning yuzini toping?



A) 10π B) $\frac{576}{7}\pi$ C) $\frac{144}{7}\pi$ D) $\frac{288}{7}\pi$

5. Hisoblang:

$$\frac{5}{6}:\frac{15}{24}+\frac{1}{2}:\frac{1}{4}$$

A) $\frac{8}{3}$ B) $\frac{10}{3}$ C) $\frac{7}{2}$ D) $\frac{11}{3}$

6. To'g'ri burchakli uchburchakning bir kateti ikkichi katetidan ikki marta ko'p. Agar uchburchak yuzi 9 cm^2 bo'lsa, kichik katetini toping.

RUSTAMBEK OQIW ORAYI

A) 4 B) 3 C) 6 D) 8

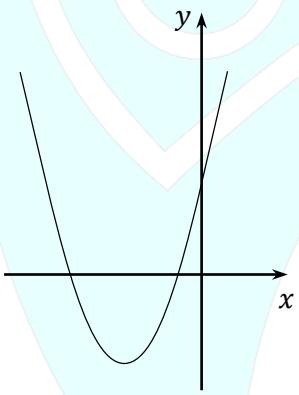
7. Yuzasi 36 ga teng bo‘lgan to’g’ri burchakli uchburchakning bir kateti orqali 360^0 ga aylantirishdan konus hosil qilindi. Konus hajmi 192π ga teng bo’lsa, konus yasovchisini toping.

A) $\sqrt{73}$ B) $\sqrt{146}$ C) 12 D) $\sqrt{145}$

8. Yuzasi 84 cm^2 ga teng bo‘lgan to’g’ri burchakli uchburchakning katta kateti orqali 360^0 ga aylantirishdan konus hosil qilindi. Konus hajmi $448\pi\ cm^3$ ga teng bo’lsa, uchburchakning katta kateti toping.

A) 13 B) 21 C) 12 D) 8

9. $f(x) = ax^2 + bx + c$ funksiya grafigi berilgan. a, b, c sonlarni taqqoslang.



A) $a > 0, b > 0, c > 0$ B) $a > 0, b < 0, c > 0$

C) $a < 0, b > 0, c < 0$ D) $a > 0, b < 0, c < 0$

10. Integralni hisoblang:

$$\int \left(\frac{1 - tg^2x}{1 + tg^2x}\right) dx$$

A) $\frac{cos2x}{2} + C$ B) $\frac{sin2x}{2} + C$ C) $tgx + C$ D) $sinx + C$

11. Aniqmas integralni hisoblang:

$$\int \frac{tgx}{\ln(cosx)} dx$$

A) $lnln(cosx) + C$ B) $-lnln(sinx) + C$

C) $-\ln\ln(\cos x) + C$ D) $lnln(tgx) + C$

12. Agar ikki sonning yig’indisi 600 ga, biri ikkinchisidan 2 marta katta bo’lsa, shu sonlardan kattasini toping.

A) 200 B) 300 C) 400 D) 450

RUSTAMBEK OQIW ORAYI

BMBA ORIGINAL SAVOLLARI

13. Aniqmas integralni hisoblang:

$$\int \frac{ctgx}{cosx}dx$$

A) $\ln tgx + C$ B) $\ln \mathrm{tg}\frac{x}{2} + C$

C) $\ln ctg\frac{x}{2} + C$ D) $\ln \cos\frac{x}{2} + C$

14. Agar $a \neq 1$ va $\int_a^1 (x+2)dx = 0$ bo'lsa, a ning qiymatini toping.

A)-5 B)-4 C)-3 D)-2

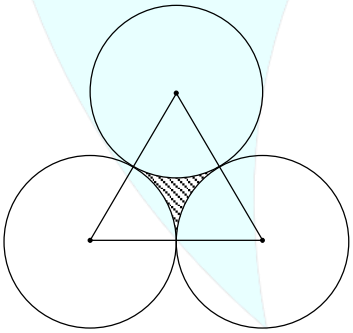
15. $\int_0^3 (|1-x| + |x+2|)dx$ ni hisoblang.

A) 18 B) 10 C) 12 D) 13

16. a, b –natural sonlar va $a + b = 13$ bo'lsa, $ab + 2024$ ning eng katta qiymatini toping.

A) 2050 B) 2066 C) 2208 D) 2025

17. Muntazam uchburchakning tomoni 8 ga teng. Bir xil radiusli aylanalar berilgan. Bo'yalgan soha yuzini toping.



A) $8\pi - 16\sqrt{3}$ B) $12\sqrt{3} - 8\pi$ C) $16\sqrt{3} - 8\pi$ D) $12\sqrt{3} - 4\pi$

18. 1, 2, 3, 4 raqamlaridan foydalanib nechta raqamlari takrorlanmaydigan uch xonali juft sonlar tuzish mumkin?

A) 16 ta B) 12 ta C) 8 ta D) 10 ta

19. Tenglama nechta ildizga ega? $10 - 2x = 2(4 - x) + 2$

A) yechimga ega emas B) Cheksiz ko'p C) 1 ta D) 2 ta

20. $f(2x) = g(x^3 - 7)$ va $g(3x + 2) = x^2 + 2x + 10$ bo'lsa, $f'(4)$ ni toping.

A) $\frac{1}{2}$ B) $\frac{1}{3}$ C) $\frac{8}{3}$ D) $\frac{2}{3}$

RUSTAMBEK OQIW ORAYI

BMBA ORIGINAL SAVOLLARI

21. Muntazam to'qqizburchakning bitta ichki burchagi necha gradusga teng?
A) 160^0 B) 140^0 C) 120^0 D) 150^0

22. $\sqrt{EKUB(18;32)\cdot EKUK(18;32)}$ ni hisoblang.
A) 576 B) 24 C) 48 D) 10

23. 6 ta bo'sh stullar bor. 2 ta o'quvchini nechta usulda shu stullarga joylashtirish mumkin?
A) 15 B) 12 C) 30 D) 60

24. $f(x)=lnx+(\ln x)^3$ funksiyaning $x=\frac{1}{e^2}$ nuqtadagi hosilasini toping.
A) $16e^2$ B) $13e^2$ C) $\frac{16}{e^2}$ D) $\frac{13}{e^2}$

25. Agar $a=7{,}5$ va $b=2{,}5$ bo'lsa, $a^3-b^3-3ab(a-b)$ ni hisoblang.
A) 120 B) 225 C) 125 D) 100

26. $\vec{a}(x;-3;1),\vec{b}(2;3;-1)$ vektorlar berilgan bu vektorlar perpendikulyar bo'lsa, x ni toping.
A) 15 B) 8 C) 5 D) 10

27. Uchburchakli prizma qirralari soni to'rtburchakli prizma qirralaridan sonidan qanchaga kam?
A) 3 B) 2 C) 4 D) 5

28. Diametrlari 6, 8, 10 ga teng bo'lgan cho'yondan ishlangan 3 ta sharsimon idish berilgan. 3 ta sharni eritib bitta katta shar quyildi. Shar diametrini toping.
A) 6 B) 8 C) 10 D) 12

29. Tengsizlikni yeching:

$$\frac{(x-1)^2\cdot\sqrt{x-2}}{(x-5)\cdot(x-6)}\geq 0$$

A) $[2;5)\cup(6;\infty)$ B) $\{1\}\cup[2;5)\cup(6;\infty)$
C) $[0;5)\cup(6;\infty)$ D) $(6;\infty)$

30. $cos3x=\sin\frac{5\pi}{6}$ tenglamaning $[0;2\pi]$ shu oraliqdagi ildizlari nechta?
A) 5 ta B) 6 ta C) 7 ta D) 4 ta

RUSTAMBEK OQIW ORAYI